

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology and Engineering

Fourth Semester of B. Tech. Examination (IT), Dec 2011

IT207: Computer Networking-I

Date: 07.12.11, Wednesday Time: 10:00a.m. To 01:00p.m. Maximum Marks: 70

Instructions:

- (1) All questions are compulsory.
- (2) Indicate clearly, the option you attempt along with its respective question Number.
- (3) Figures to right indicate marks.
- (4) Rough work is done in the last page of main supplementary; don't write anything on the question paper.

SECTION-I

Q-1 Answer the following questions. [11]

- [A] A signal has been received that only has values of -1, 0 and 1. Is this an analog signal or digital signal? [1]
- [B] Name the five Basic networks Topology. [1]
- [C] _____ type of plot/diagram shows the phase and amplitude of signal at give time. [1]
- [D] How do the layers of the TCP/IP protocol suite correlate to the layers of the OSI model? [2]
- [E] How does NRZ-L differ from NRZ-I? [2]
- [F] What are the advantages and disadvantages of parallel transmission? [2]
- [G] What are the functions of a DTE and DCE? Give an example of each. [2]

Q-2 Answer the following questions. [12]

- [A] Write a short note on Pulse Code Modulation (PCM). [6]
- [B] What is purpose OSI model? Write down functionalities of each layer of OSI model. [6]

OR

- [A] Write down the differences between OSI model and TCP/IP model. [4]
- [B] Define the three Transmission modes of communication. [4]
- [C] Write down electrical, mechanical and functional characteristics of EIA-232 standard. [4]

Q-3 Answer the following questions.(Attempt any Three) [12]

- [A] What are the advantages of distributed processing?
- [B] Explain Amplitude shift keying.
- [C] B taking suitable example, explain B8ZS encoding technique.
- [D] Write a short note on: 56Kbps Modem.

SECTION-II

- Q-4 Answer the following questions. [11]**
- [A] What is an advantage of coaxial cable over twisted pair cable? [1]
 - [B] How is CRC superior than LRC? [1]
 - [C] How do guided media differ from unguided media? [1]
 - [D] What is purpose of line discipline? [1]
 - [E] HDLC is a _____ protocol. [1]
1. Character Oriented 2. Bit-Oriented 3. Byte-oriented 4. Count-Oriented
 - [F] What are the two sublayers of Data link Control layer? What are the responsibilities of sublayers? [2]
 - [G] What is data transparency in BSC? [2]
 - [H] What are throughput, Propagation speed and propagation time in data communication network? [2]
- Q-5 Answer the following questions. [12]**
- [A] What is Time Division Multiplexing? By taking diagram example, explain two types of TDM. [6]
 - [B] Write a short note on : HDLC [6]
- OR**
- [A] Write a short note on: Packet Switching [6]
 - [B] Write short Note on: Types of Propagation of Radio Waves [6]
- Q-6 Answer the following questions.(Attempt any Three) [12]**
- [A] By taking suitable, explain CRC.
 - [B] Explain CSMA/CD mechanism.
 - [C] What is mechanism of sliding window protocol in flow control?
 - [D] Write down the media access mechanism in DQDB.